

Frontend Integration Class Diagram

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Dokumen ini fokus pada diagram kelas dengan atribut per class, dikelompokkan berdasarkan data layer, workflow layer, dan helper layer.

1. Class Diagram Inti

```
```mermaid
```

```
classDiagram
```

```
class User {
 +id: bigint
 +first_name: string
 +last_name: string
 +phone: string
 +email: string
 +username: string?
 +password: hashed
 +role: string
 +is_active: bool
 +last_login_at: datetime?
 +full_name: string
}
```

```
class FarmerProfile {
 +id: bigint
 +user_id: bigint
 +full_name: string
 +role_label: string
 +location: string
 +village: string?
```

```
+district: string?
+city: string?
+contact: string?
+avatar_path: string?
}
```

```
class CollectorProfile {
 +id: bigint
 +user_id: bigint
 +business_name: string
 +address: string?
 +contact: string?
 +avatar_path: string?
}
```

```
class DistributorProfile {
 +id: bigint
 +user_id: bigint
 +business_name: string
 +address: string?
 +contact: string?
 +avatar_path: string?
}
```

```
class UmkmProfile {
 +id: bigint
 +user_id: bigint
 +name: string
```

```
+owner_name: string
+about: string?
+address: string?
+contact: string?
+image_path: string?
}
```

```
class ConsumerProfile {
 +id: bigint
 +user_id: bigint
 +display_name: string
 +address: string?
 +phone: string?
 +avatar_path: string?
}
```

```
class Farm {
 +id: bigint
 +farmer_user_id: bigint
 +name: string
 +province: string?
 +city: string?
 +district: string?
 +village: string?
 +address: string?
 +latitude: decimal(7)?
 +longitude: decimal(7)?
 +notes: text?
```

```
}
```

```
class HarvestBatch {
 +id: bigint
 +code: string
 +farmer_user_id: bigint
 +farm_id: bigint
 +farm_name_snapshot: string
 +variety: string?
 +grade: string?
 +quantity_kg: decimal(2)?
 +unit: string?
 +fruit_count: int?
 +harvest_date: date?
 +status: string
 +fertilizer: string?
 +harvest_method: string?
 +maturity_level: string?
 +shelf_life_estimate: string?
 +storage_suggestion: string?
 +notes: text?
 +photo_path: string?
 +received_quantity_kg: decimal(2)?
 +received_fruit_count: int?
 +verified_grade: string?
 +quality_notes: text?
 +verified_by_user_id: bigint?
 +verified_at: datetime?
```

```
+rejection_reason: text?
+rejected_by_user_id: bigint?
+rejected_at: datetime?
+blockchain_network: string?
+blockchain_status: string?
+blockchain_anchor_ref: string?
+blockchain_tx_hash: string?
+blockchain_block_number: int?
+blockchain_payload: json?
+blockchain_synced_at: datetime?
}
```

```
class HarvestBatchEvent {
 +id: bigint
 +harvest_batch_id: bigint
 +title: string
 +actor_label: string?
 +actor_user_id: bigint?
 +event_at: datetime
 +metadata: json
 +previous_ledger_hash: string?
 +ledger_hash: string?
 +ledger_height: int
}
```

```
class HarvestBatchGradeBreakdown {
 +id: bigint
 +harvest_batch_id: bigint
```

```
+grade_label: string
+weight_kg: decimal(2)
+fruit_count: int
}
```

```
class CollectorShipment {
 +id: bigint
 +code: string
 +collector_user_id: bigint
 +destination_type: string
 +total_weight_kg: decimal(2)?
 +total_fruit_count: int?
 +packaged_at: datetime?
 +warehouse_note: text?
 +sent_at: datetime?
 +completed_at: datetime?
 +status: string
}
```

```
class CollectorShipmentSource {
 +id: bigint
 +collector_shipment_id: bigint
 +harvest_batch_id: bigint
 +source_code_snapshot: string
 +source_grade_snapshot: string?
 +source_weight_kg: decimal(2)?
 +source_fruit_count: int?
 +source_variety_snapshot: string?
```

```
}
```

```
class DistributorReceipt {
 +id: bigint
 +code: string
 +distributor_user_id: bigint
 +collector_shipment_id: bigint
 +expected_weight_kg: decimal(2)?
 +expected_fruit_count: int?
 +received_weight_kg: decimal(2)?
 +received_fruit_count: int?
 +condition: string?
 +received_at: datetime?
 +discrepancy_note: text?
 +quality_note: text?
}
```

```
class UmkmProduct {
 +id: bigint
 +code: string
 +umkm_user_id: bigint
 +category: string?
 +name: string
 +price: decimal(2)?
 +stock_qty: int?
 +description: text?
 +status: string
 +photo_path: string?
```

```
}
```

```
class UmkmProductSource {
 +id: bigint
 +umkm_product_id: bigint
 +harvest_batch_id: bigint
 +source_code_snapshot: string
 +weight_kg: decimal(2)?
 +fruit_count: int?
}
```

```
class UmkmOrder {
 +id: bigint
 +code: string
 +umkm_product_id: bigint
 +buyer_name: string
 +buyer_phone: string?
 +buyer_address: string
 +quantity: int
 +total_amount: decimal(2)?
 +qr_code_data: string?
 +note: text?
 +status: string
}
```

```
class ConsumerTransaction {
 +id: bigint
 +code: string
```



```
+consumer_user_id: bigint
+umkm_product_id: bigint
+product_name_snapshot: string?
+quantity: int
+total_amount: decimal(2)?
+buyer_coordinates: string?
+payment_status: string
+status: string
+qr_code_data: string?
+note: text?
+paid_at: datetime?
+processed_at: datetime?
}
```

```
class ApiResponse {
 +success(data, message, status)
 +created(data, message)
 +error(message, status, errors)
 +validation(errors, message)
}
```

```
class ContractFormatter {
 +user(user)
 +farm(farm)
 +batch(batch)
 +batchEvent(event)
 +batchTimeline(batch)
 +shipment(shipment)
```

```
+receipt(receipt)
+product(product)
+order(order)
+transaction(transaction)
+trace(batch, publicUrl)
+moneyLabel(value)
}
```

```
class DummyBlockchainLedger {
 +NETWORK: string
 +STATUS: string
 +proof(batch)
 +sync(batch)
 +prepareEvent(batch, title, actor, metadata, actorLabel)
 +eventProof(batch, event)
}
```

```
class CodeGenerator {
 +batch(id)
 +shipment(id)
 +receipt(id)
 +product(id)
 +order(id)
 +transaction(id)
}
```

```
class Traceability {
 +recordEvent(batch, title, actor, metadata, actorLabel)
```

}

User "1" --> "0..1" FarmerProfile : hasOne

User "1" --> "0..1" CollectorProfile : hasOne

User "1" --> "0..1" DistributorProfile : hasOne

User "1" --> "0..1" UmkmProfile : hasOne

User "1" --> "0..1" ConsumerProfile : hasOne

User "1" --> "many" HarvestBatch : farmer

Farm "1" --> "many" HarvestBatch : harvestBatches

HarvestBatch "1" --> "many" HarvestBatchEvent : events

HarvestBatch "1" --> "many" HarvestBatchGradeBreakdown : gradeBreakdowns

User "1" --> "many" CollectorShipment : collector

CollectorShipment "1" --> "many" CollectorShipmentSource : sources

CollectorShipmentSource "\*" --> "1" HarvestBatch : batch

User "1" --> "many" DistributorReceipt : distributor

DistributorReceipt "1" --> "1" CollectorShipment : shipment

User "1" --> "many" UmkmProduct : umkm

UmkmProduct "1" --> "many" UmkmProductSource : sources

UmkmProductSource "\*" --> "1" HarvestBatch : batch

UmkmProduct "1" --> "many" UmkmOrder : orders

UmkmProduct "1" --> "many" ConsumerTransaction : transactions

ConsumerTransaction "\*" --> "1" User : consumer

ContractFormatter ..> HarvestBatch : serialize

ContractFormatter ..> CollectorShipment : serialize

ContractFormatter ..> UmkmProduct : serialize

DummyBlockchainLedger ..> HarvestBatch : proof/sync

Traceability ..> HarvestBatch : record events

...

## 2. Catatan Fokus

Diagram ini menonjolkan atribut yang memang dipakai backend sebagai sumber data utama, termasuk field snapshot, status, ledger, dan relasi antarmodel.

- Data master identitas: User dan seluruh profile turunan.
- Traceability inti: HarvestBatch, HarvestBatchEvent, HarvestBatchGradeBreakdown.
- Distribusi: CollectorShipment, CollectorShipmentSource, DistributorReceipt.
- Produk dan transaksi: UmkmProduct, UmkmProductSource, UmkmOrder, ConsumerTransaction.
- Helper lintas alur: ApiResponse, ContractFormatter, DummyBlockchainLedger, CodeGenerator, Traceability.